ELE1204: Digital Electronics and Logic Design

Lecture 03 (BSCS)

Topic/Theme, e.g., *Combinational Logic Design*

Ms. Olivia Nakayima
(MSc, BSc. CCNA, JNCIA, JNCIS)
Department of Computing & Technology
Faculty of Engineering, Design & Technology

Mon 16th October 2023

Lecture Objectives and Learning outcomes

The Objectives of this lecture are :

- To understand the combinational circuits
- To analyse them and their applications

By the end of this lecture, students should be able to:

- Define combinational circuits
- Fully explain the adders, multiplexers, encoders and decoders

NOTE:: SMART = **S**pecific, **M**easurable, **A**chievable, **R**elevant, **T**imebound

Lecture Overview

- Introduction and characteristics
- Analysis of basic combinational circuits
- Adders (Half, Full), Subtractors
- Multiplexers (MUX) and Demultiplexers (DEMUX)
- Encoders and Decoders, Code Converters Parity Generators and Checkers

Multiplexers (MUX) and Demultiplexers (DEMUX)

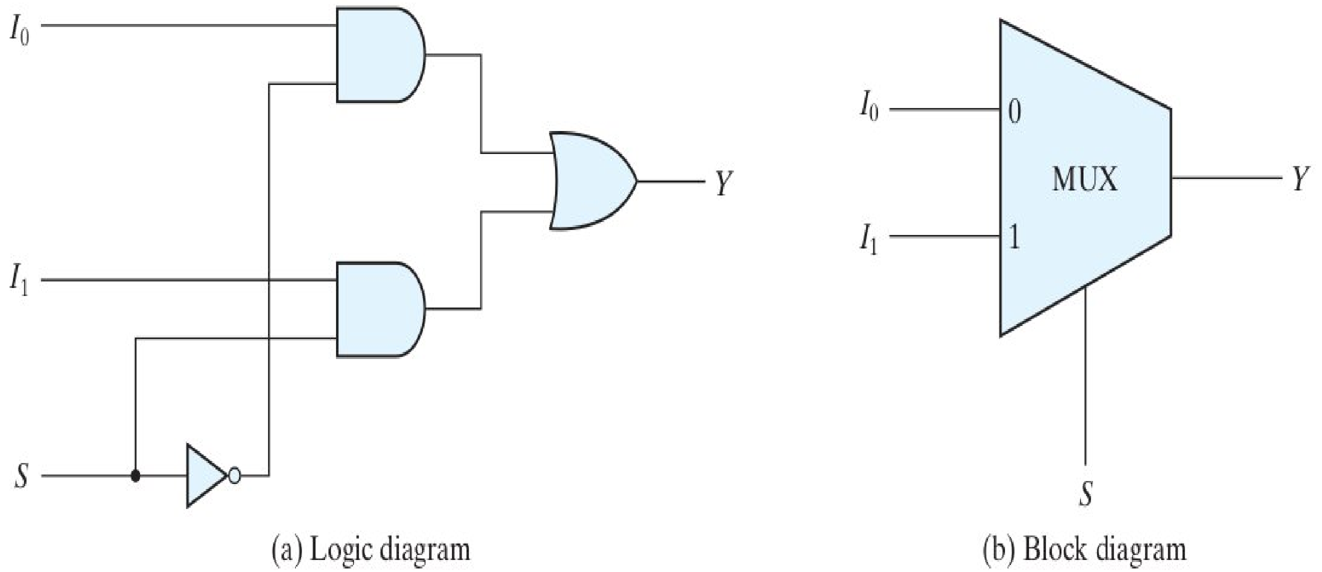
Multiplexer (MUX)

2- to- 1 multiplexer
https://www.youtube.com/watch?v=FKvnmxte98A

- A **MUX** selects **one** output from **multiple inputs** based on **select lines**.

Example:

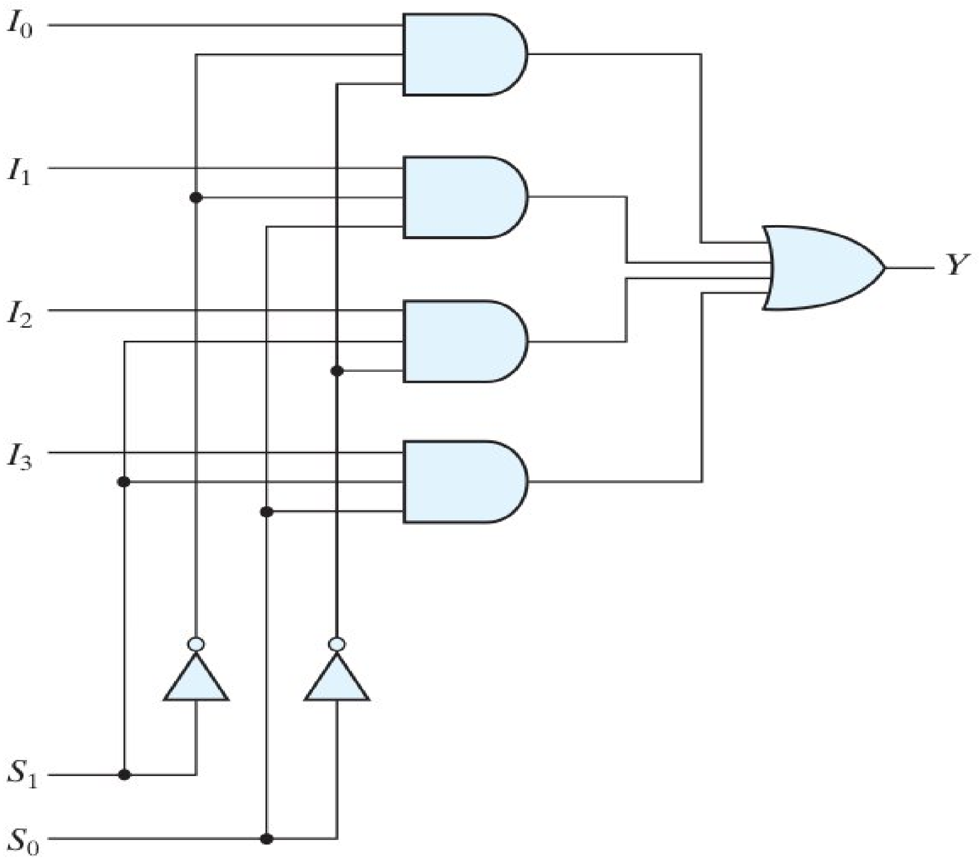
- **2-to-1 MUX** has **2 inputs (I_0, I_1)**, **1 select line (S)**, and **1 output (Y)**.
- When S = 0, the upper AND gate is enabled and has a path to the output. When S = 1, the lower AND gate is enabled and I 1 has a path to the output.
- The multiplexer acts like an electronic switch that selects one of two sources.
- A multiplexer is also called a data selector , since it selects one of many



Multiplexers (MUX) and Demultiplexers (DEMUX)

4- to- 1 multiplexer

- Each of the four inputs, I_0 through I_3, is applied to one input of an AND gate.
- Selection lines S 1 and S 0 are decoded to select a particular AND gate.
- The outputs of the AND gates are applied to a single OR gate that provides the one-line output



(a) Logic diagram

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

(b) Function table

https://www.youtube.com/watch?v=g1Lfz1XgrH8

Multiplexers (MUX) and Demultiplexers (DEMUX)

8 input multiplexer

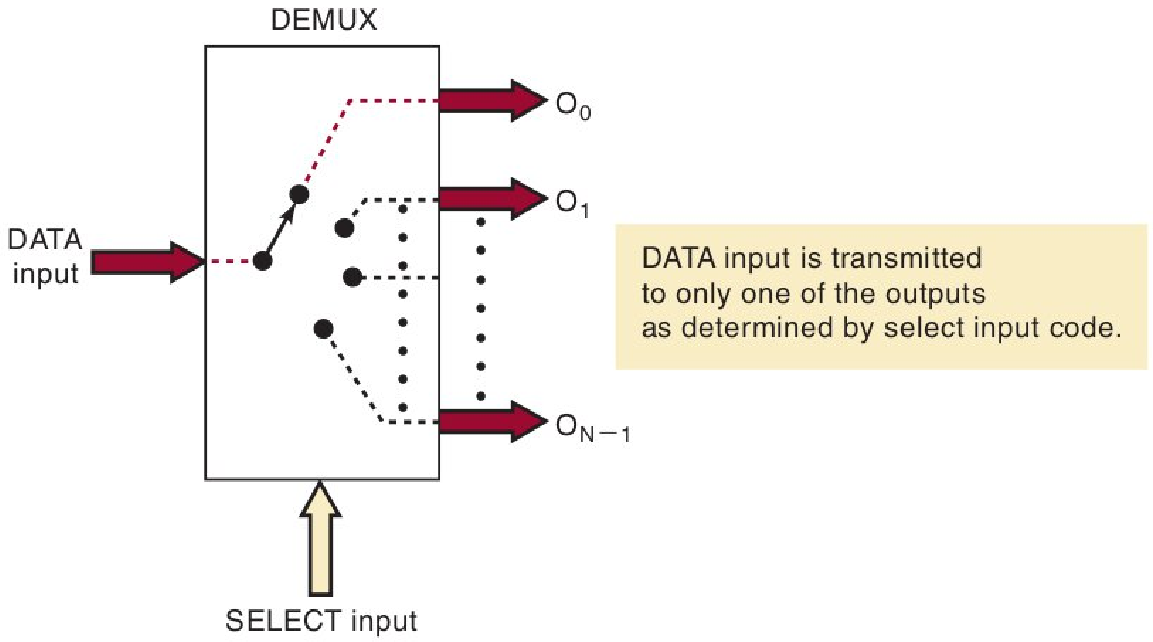
- Has 3 selectors
- 8 input lines
- Read about the 8 bit MUX

- *https://www.youtube.com/watch?v=b0z7YKKCCyY*

Multiplexers (MUX) and Demultiplexers (DEMUX)

Demultiplexer (DEMUX)

- A **DEMUX** takes **one input** and distributes it to **multiple outputs** based on select lines.
- Demultiplexer takes one input data source and selectively distributes it to 1 of N output channels just like a multiposition switch
- The select input code determines to which output the DATA input will be transmitted
- ***https://www.youtube.com/watch?v=t3Ed13z9uz8***



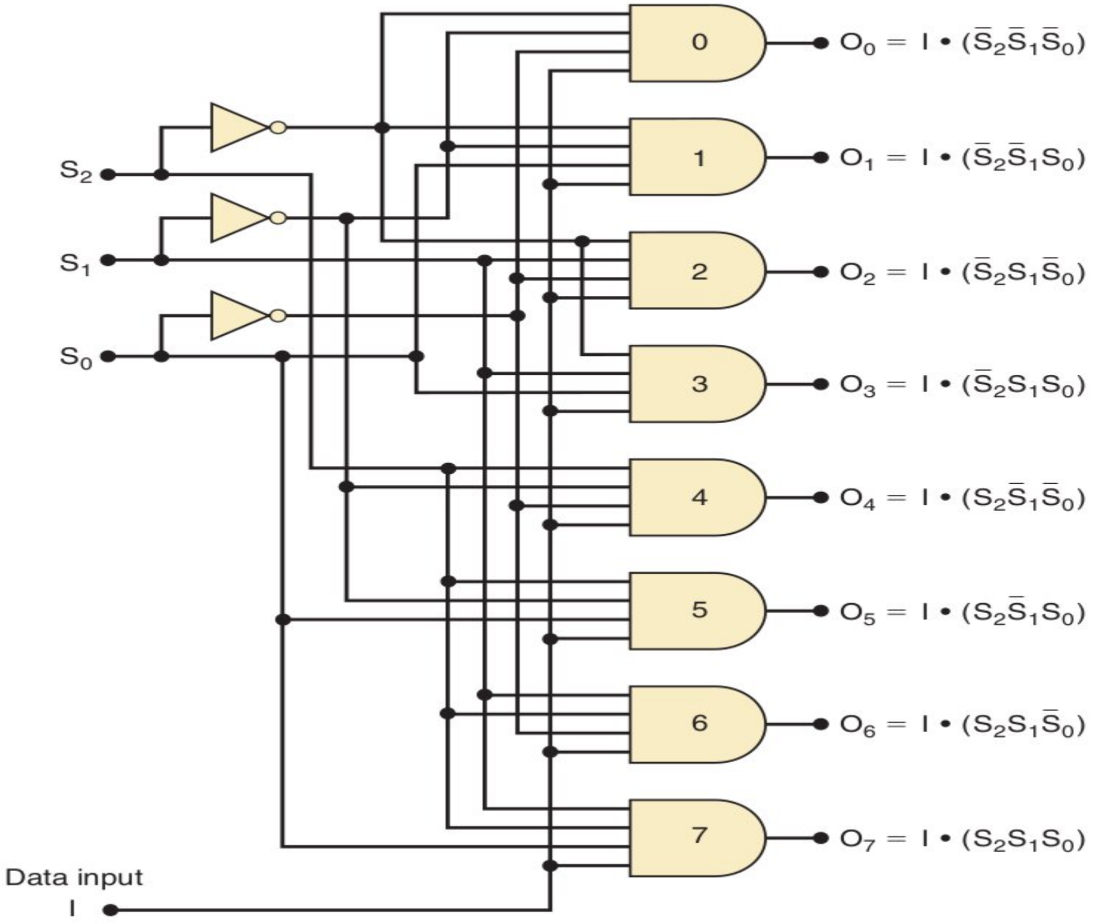
Used in **data routing and memory addressing**.

Multiplexers (MUX) and Demultiplexers (DEMUX)

1-line to 8-line Demultiplexer (DEMUX)

Select Code			Outputs							
S_2	S_1	S_0	O_7	O_6	O_5	O_4	O_3	O_2	O_1	O_0
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0

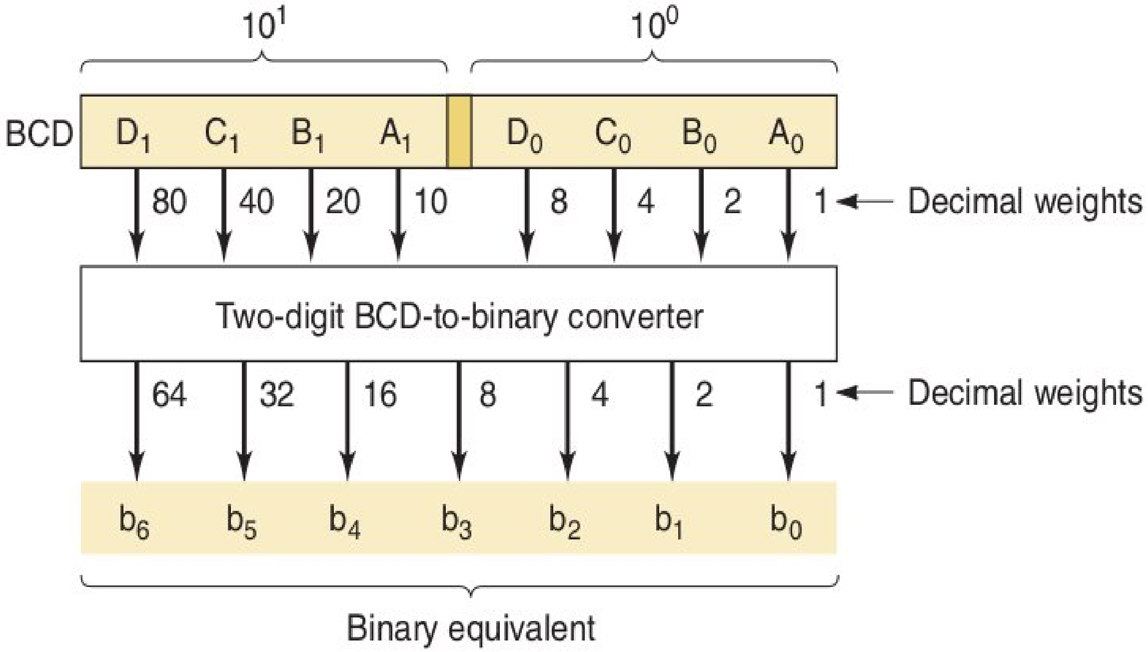
Note: I is the data input



Code Converters, Parity Generators, and Checkers

Code Converters

- A code converter is a logic circuit that changes data presented in one
- type of binary code to another type of binary code
- **Example: Binary to BCD, Binary to Gray Code.**



Code Converters, Parity Generators, and Checkers

Code Converters example

Convert 01010010 (BCD for decimal 52) to binary. Repeat for 10010101 (decimal 95).

Solution

Write down the binary equivalents for all the 1s in the BCD representation. Then add them all together in binary.

```
0 1 0 1 0 0 1 0    (BCD)
                    0000010 (binary for 2)
                    0001010 (binary for 10)
                  + 0101000 (binary for 40)
                    0110100 (binary for 52)
1 0 0 1 0 1 0 1    (BCD)
                    0000001 (binary for 1)
                    0000100 (binary for 4)
                    0001010 (binary for 10)
                  + 1010000 (binary for 80)
                    1011111 (binary for 95)
```

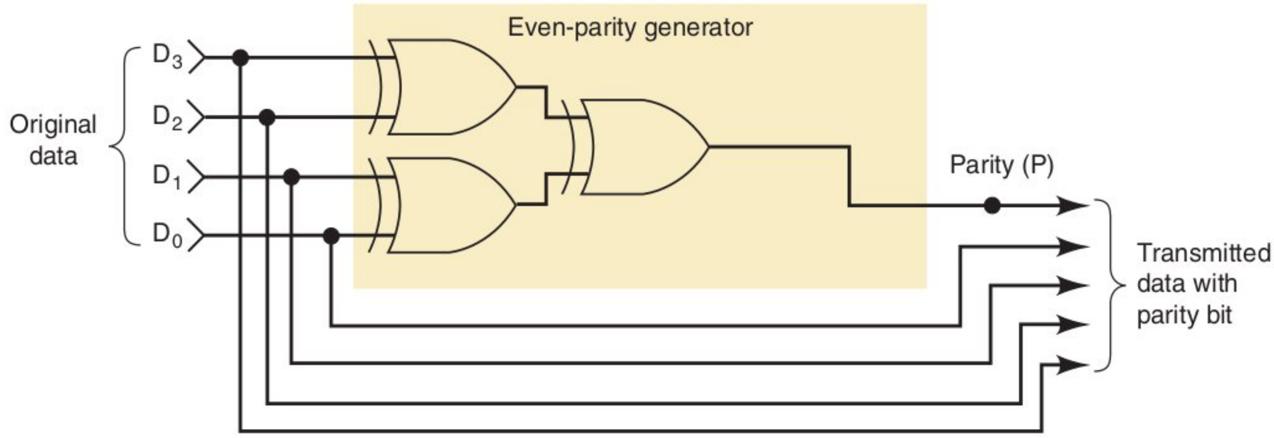
https://www.youtube.com/watch?v=cF-Q5j7RUEw

Code Converters, Parity Generators, and Checkers

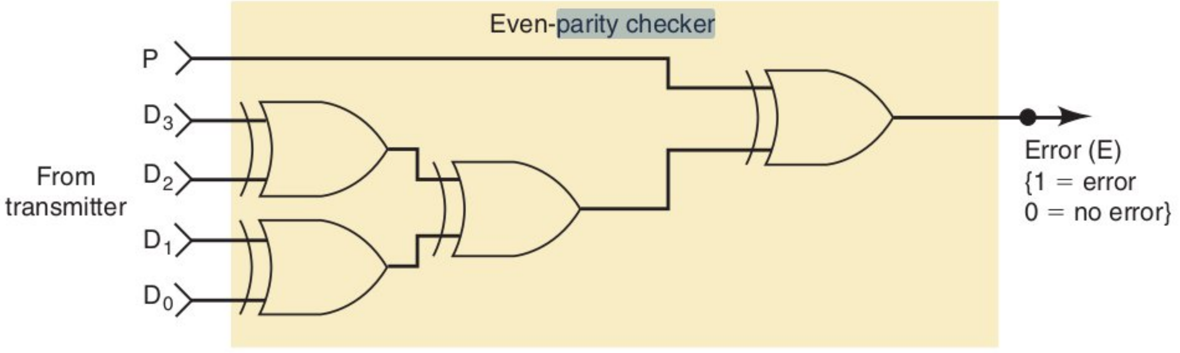
Parity Generators and Checkers

- A transmitter can attach a parity bit to a set of data bits before transmitting the data bits to a receiver.
  - **Parity bit** is used for **error detection** in data transmission.
  - **Even parity**: Ensures the number of 1s is even.
  - **Odd parity**: Ensures the number of 1s is odd.

https://www.youtube.com/watch?v=c8qAti1zBVQ



(a)



(b)

P.O. Box 4, Mukono, Uganda, Plot 67-173, Bishop Tucker Road, Mukono Hill | Tel: +256 (0) 312 350 800 Email: info@ucu.ac.ug Web: https://ucu.ac.ug

Thank You

Uganda Christian University
P.O. Box 4 Mukono, Uganda
Tel: 256-312-350800
https://ucu.ac.ug/ Email: info@ucu.ac.ug.
@ugandachristianuniversity @UCUniversity
@UgandaChristianUniversity

Department of Computing & Technology
FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
Tel: +256 (0) 312 350 863 | WhatsApp: +256 (0) 708 114 300
@ucucomputeng @ucu_ComputEng
https://cse.ucu.ac.ug/ Email: dct-info@ucu.ac.ug